

HMR-BiLSTM: A Robust and Interpretable Hybrid Memory Residual BiLSTM Framework for ECG Arrhythmia Classification

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Abstract. Automatic ECG arrhythmia classification is now common in cardiac monitoring, but detecting rare and dangerous beat types is still difficult. LSTM-based models are popular for this task, yet they struggle on several fronts: long-range dependencies across 187-sample heartbeats, class imbalance that reaches $113\times$, and no built-in way for a clinician to see what the model is doing. We present HMR-BiLSTM, a BiLSTM augmented with a Residual Memory Control (RMC) cell and a Hybrid Memory Path. The RMC cell adds an interaction-aware residual gate on top of the standard LSTM gates and splits memory into keep and add streams weighted by learned softmax attention. Attention pooling weights, computed after the recurrent loop, provide a signal that can be overlaid on the ECG waveform for inspection. The Hybrid Memory Path runs the RMC cell alongside a conventional LSTM cell through a learnable blend gate, so the model behaves like a standard LSTM early in training and relies on the RMC pathway more as training progresses. On MIT-BIH, the model reaches macro AUC 0.9916, macro Recall 0.9146, and macro F1 0.8825, with recall 4.43 percentage points above CNN-BiLSTM-ATTN and gains concentrated on the Fusion and Supraventricular classes. Under FGSM at $\epsilon=0.02$, HMR-BiLSTM holds macro F1 at 0.842 versus 0.773 for CNN-BiLSTM-ATTN; under PGD-20 it holds at 0.839 versus 0.756. Across these tests, HMR-BiLSTM holds up better than the recurrent baselines, though we stop short of any deployment claim.

Keywords: ECG arrhythmia classification, Interpretable AI, Robust medical AI, Bidirectional LSTM, Residual Memory Control, Adversarial robustness, AI Safety.

1 Introduction

Automatic heartbeat classification from ECG signals now runs across a wide range of devices, from Holter monitors to smartwatches, and many of these systems are built on a basic LSTM. That backbone carries three weaknesses. First, its forget gate multiplies element-wise, which leaves it unable to capture nonlinear interactions

between the cell state and the hidden state. Second, its gating drifts across 187 time steps once Normal beats outnumber the rarest class by 113 to 1. Third, nothing inside the cell tells a clinician why a given beat was flagged. None of these is a minor issue. A missed ventricular beat can delay treatment, so a fair evaluation has to look past leaderboard accuracy: the model also has to stay steady when the signal is noisy and give cardiologists something they can actually read. Adversarial robustness adds a further requirement, rarely tested in ECG work but hard to ignore once the output feeds a clinical decision.

We add a Residual Memory Control (RMC) mechanism inside a standard BiLSTM cell, giving HMR-BiLSTM. The cell combines an interaction-aware gate, a keep/add memory decomposition, and softmax attention over both streams. In early trials, replacing the LSTM update entirely with RMC destabilised training, so we keep both pathways through a Hybrid Memory Path and let a learnable blend gate weigh their contributions. We initialise the blend gate's bias at -5 , which keeps the RMC branch mostly off in early epochs and opens it up as training progresses. On MIT-BIH (AAMI EC57 protocol), macro AUC reaches 0.9916 against 0.9803 for CNN-BiLSTM-ATTN, and macro recall rises from 0.8703 to 0.9146. The attention pooling weights, available at every time step, act as the clinician-facing signal; the residual gate works mainly as internal memory routing. We also run FGSM and PGD attacks and a calibration analysis.

2 Related Work

ECG classification is a multi-class time-series problem made harder by severe class imbalance [4]. Earlier work leaned on hand-extracted features like the RR interval, QRS width, and P/T-wave amplitude, fed to a Logistic Regression [1] or Decision Tree [2]. These are easy to follow but break under noise or population shift. Deep models dropped that step: on MIT-BIH, 1D CNNs [5] and BiLSTMs both do well, with Yildirim [6] hitting 99.39% accuracy and Hannun et al. [7] matching cardiologists across 91,000 patients. Attention [8] and Transformer blocks [15] later sharpened the focus on diagnostically relevant segments, but the recurrent core itself was left largely untouched. Greff et al. [9] mapped the LSTM cell's design space, and Zoph and Le [10] found searched structures that outperform it, which is where the memory-augmented cell we develop here comes in.

Once these models reach the clinic, explaining them is no longer optional. SHAP [11] and LIME [12] rank input-feature importance but cannot show how memory shifts across the P-wave, QRS complex, and T-wave. Attention maps [15] show where the model looks, not the memory dynamics inside the cell. Rudin [13] argues that high-stakes medicine should use models interpretable by design, not explained after the fact, and clinical-readiness surveys keep returning to the same shortlist: reliability, transparency, robustness, calibration [16,17].

And there is nothing hypothetical about that robustness. Han et al. [18] evaded both CNN and RNN ECG classifiers with minuscule perturbations, while Begoli et al. [19] cautioned that a model with no sense of its own uncertainty should not be

deployed on its own. The gap is consistent: memory-augmented networks add capacity but expose nothing a clinician can inspect, while post-hoc tools sit outside the model and miss what happens in the recurrent loop. HMR-BiLSTM addresses this with a lightweight mechanism inside the LSTM cell, paired with attention pooling weights that can be plotted next to the ECG signal for inspection.

3 Proposed Methodology

Category	Component	Role / Limitation
Existing	Standard LSTM [3,9]	Multiplicative forget gate; no nonlinear c/h interaction under class imbalance
Existing	Attention [8,15]	Weights on hidden states; cannot track cell memory across P-QRS-T phases
Existing	SHAP/LIME [11,12]	Post-hoc black-box attribution; may misrepresent internal gate dynamics [13]
Existing	Residual RNN variants [9,10]	External capacity modules; no clinician-readable signal inside recurrent loop
Proposed	RMC Cell	Interaction-aware gate r_t with keep/add memory decomposition via softmax attention
Proposed	Hybrid Memory Path	Parallel LSTM+RMC branches via blend gate β_t (init ≈ 0.007); gradient-driven transition
Proposed	Temporal Smoothness ($\lambda=0.003$)	Penalises abrupt gate shifts; stabilises temporal memory routing during training
Proposed	Attention Pooling	Adaptive weights γ_t over $T=46$ steps; complements but does not replace RMC gates

Scope of Contribution. HMR-BiLSTM does not reinvent the recurrent cell; the foundational LSTM equations remain intact. The paper introduces a specialised memory routing mechanism — the RMC cell and Hybrid Memory Path — that mitigates two failure modes of conventional LSTMs on long ECG sequences: poor minority-class recall under extreme imbalance and the absence of intrinsic time-step interpretability. The smoothness constraint and attention pooling serve as stabilisation and temporal focus layers respectively; the core novelty resides in the hybrid blending mechanism, which trades a modest parameter increase for improved minority-class recall and built-in inspectability on safety-critical cardiac signals. Robustness gains reported in Section 5 reflect the combined effect of the RMC architecture and adversarial training; their individual contributions are separated in the ablation (Section 5.2).

3.1 Problem Formulation

Given an input ECG segment $X = (x_1, x_2, \dots, x_T)$ with $x_t \in \mathbb{R}^d, \forall t \in \{1, 2, \dots, T\}$ with $T = 187$ and $d = 1$ for single-lead MIT-BIH signals, the goal is to learn a function $f: \mathbb{R}^{T \times d} \rightarrow \Delta^{K-1}$ that maps each segment to a probability distribution over $K = 5$ heartbeat classes following the AAMI EC57 standard [21] (N, S, V, F, Q).

Training uses Focal Loss with per-class weights:

$$\mathcal{L}_{\text{task}} = -\frac{1}{N} \sum_i \sum_k w_k \cdot \mathbf{1}[y_i = k] \cdot (1 - p_{i,k})^\gamma \cdot \log p_{i,k}, \gamma = 1.5$$

We compute the modulating term $(1 - p_{i,k})^\gamma$ from the unweighted cross-entropy before applying the per-class weight w_k . This ordering differs from the original formulation of Lin et al. [24], where α is folded into the focal term directly. Our variant **is designed to make** the focusing mechanism $(1 - p_t)^\gamma$ **reflect** the model's raw prediction difficulty, independent of the class prior encoded in w_k . In practice the difference is small, but we note it for reproducibility. Class weights follow the scikit-learn balanced formula and are clipped to $[0.5, 10.0]$.

3.2 HMR-BILSTM Architecture

The architecture consists of three stacked blocks. The CNN frontend applies two Conv1D–BN–ReLU–MaxPool blocks (kernel sizes 5 and 3, pooling stride 2) and reduces the sequence length from $T = 187$ to $T' = 46$ while expanding the channel dimension from 1 to 64, designed to capture local waveform patterns before they enter the recurrent block. The HMR-BILSTM block stacks two layers of bidirectional LSTM Cells (num_layers = 2), where the hidden output of the first layer serves as the input sequence to the second. Each cell uses the Residual Memory Control mechanism combined with a Hybrid Memory Path (Sections 3.3–3.4), and the final layer's hidden states are concatenated across directions to form a sequence of dimension $2H = 192$ before attention pooling. Finally, an attention pooling layer learns adaptive weights γ_t over time steps and aggregates the hidden states, after which a two-layer fully-connected classifier with Layer Normalization and dropout produces the class logits. The full model has approximately 505K trainable parameters.

3.3 Residual Memory Control Cell

A standard LSTM updates its cell memory as $c_t = f_t \odot c_{t-1} + i_t \odot g_t$. The forget gate is a simple element-wise multiplication and cannot capture nonlinear interactions between c_{t-1} and h_{t-1} when deciding what to keep. Our HMR cell adds a small interaction-aware residual gate:

$$r_t = \sigma(\text{LayerNorm}(W_c c_{t-1} + W_h h_{t-1} + m_t)), m_t = (W_c c_{t-1}) \odot (W_h h_{t-1})$$

Intuitively, r_t provides a per-dimension adjustment to the forget gate's keep/add balance. The memory is then split into a "keep" stream $c_t^{\text{keep}} = (f_t + r_t \odot (1 - f_t)) \odot c_{t-1}$ and an "add" stream $c_t^{\text{add}} = (1 - r_t) \odot (i_t \odot g_t)$, which are combined

through a learned softmax attention. Before computing the softmax scores, both c_keep and c_add are layer-normalised to prevent either stream from dominating by scale alone. The two streams are then weighted according to their joint content rather than raw memory values.

3.4 Hybrid Memory Path and Temporal Smoothness

Replacing the standard memory update entirely with RMC carries two risks: slow convergence early in training, and stream collapse when the softmax attention saturates. To avoid both risks, we run the RMC pathway c_t^{rmc} in parallel with the standard LSTM pathway c_t^{LSTM} and merge them through a learnable blend gate:

$$\beta_t = \sigma(W_\beta h_{t-1}), c_t = \beta_t \odot c_t^{rmc} + (1 - \beta_t) \odot c_t^{LSTM} \quad (5)$$

The bias of W_β is initialized to -5 , giving $\beta_t = \sigma(-5) \approx 0.007$ at the start of training. At initialisation, then, the RMC branch is nearly shut off. The network starts training as a conventional LSTM and only ramps up the residual memory channel if and when the loss gradients push it to.

Because ECG is a continuous signal — not a bag of independent tokens — we want r_t to change smoothly from one time step to the next. We enforce this with a smoothness penalty:

$$\mathcal{L} = \mathcal{L}_{task} + \frac{\lambda}{T-1} \sum_{t=2}^T \|r_t - r_{t-1}\|_2^2, \lambda = 0.003 \quad (6)$$

The smoothness penalty stabilises the gate dynamics during training, discouraging abrupt changes in r_t between adjacent time steps so that memory routing evolves gradually across the cardiac cycle.

4 Experimental Protocol

4.1 Dataset, Baselines, and Models

All experiments use the MIT-BIH Arrhythmia Database [20] in the preprocessed Kaggle version released by Fazeli. A stratified 85/15 split yields 74,420 training and 13,134 validation samples, with a held-out test set of 21,892. Each beat is fixed at $T = 187$ steps and standardised via Z-score normalisation fit on the training set. The five AAMI EC57 classes are distributed as N (82.77%), S (2.54%), V (6.61%), F (0.73%), and Q (7.35%), giving a $113\times$ imbalance.

We compare six methods: two classical baselines (Logistic Regression and Decision Tree), three deep baselines (CNN-LSTM-ATTN, CNN-BiLSTM-ATTN, and ResNet1D, a four-block 1D residual network representing a modern CNN baseline [5,14]), and the proposed HMR-BiLSTM. All deep models share the same CNN frontend and attention pooling, differing only in the recurrent core. Comparison is limited to architectures of comparable scale (200K–510K parameters); Transformer models are an orthogonal direction left for future work.

All deep models train with Adam ($\text{lr} = 1 \times 10^{-3}$, weight decay 1×10^{-4}), batch size 128, dropout 0.25, and hidden size 96, with seed 42. HMR-BiLSTM runs up to 45 epochs with cosine annealing to 1×10^{-5} , patience 8 on validation macro F1, gradient clipping 1.0, and Focal Loss ($\gamma = 1.5$). The baselines use a fixed learning rate, up to 12 epochs, patience 4, gradient clipping 0.5, and weighted cross-entropy with the same class weights. This split is intentional: Focal Loss and cosine annealing are part of HMR-BiLSTM's design, while the baselines follow their conventional configuration. HMR-BiLSTM also incorporates lightweight adversarial training: at each step, 30% of the batch is replaced with FGSM examples at $\epsilon = 0.02$. The baselines train on clean data only.

4.2 Security Threat Model

We assume a white-box adversary who has the model weights and can craft bounded L_∞ perturbations on the ECG input but cannot retrain the model. The threat tested is input perturbation to flip predictions, evaluated under FGSM and PGD (Section 5.3) and Gaussian noise (Section 5.4). Three additional threats (data poisoning, model extraction, and membership inference on the gate trajectories) are acknowledged but not tested.

4.3 Metrics and Research Hypotheses

Classification is evaluated with Accuracy, macro Precision/Recall/F1, weighted F1, and macro AUC (One-vs-Rest), with macro metrics as primary since they weight all classes equally — matching the clinical priority of detecting rare arrhythmias. Calibration uses ECE and Brier score. Robustness is tested under adversarial attacks (FGSM and PGD, $\epsilon \in \{0, 0.01, 0.02, 0.05\}$, and additive Gaussian noise ($\sigma \in \{0, 0.01, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5\}$). The experiments test three hypotheses. **H1**: sequential models outperform classical classifiers when the input has temporal structure. **H2**: HMR-BiLSTM improves over CNN-BiLSTM-ATTN, with the Residual Memory Control and blend gate as the hypothesised source. **H3**: HMR-BiLSTM improves detection of rare classes and degrades more gracefully under adversarial and natural perturbations.

5 Results and Discussion

5.1 Overall Comparison

Table 1. Performance on MIT-BIH ECG test set (21,892 samples). Column maximums are bolded.

Method	Acc	P_mac	R_mac	F1_ma c	F1_w	AUC
Logistic Regression	0.6743	0.4408	0.7616	0.4787	0.7243	0.9145
Decision Tree	0.8911	0.5812	0.8135	0.6846	0.8890	0.9197
CNN-LSTM-	0.9715	0.8665	0.8734	0.8691	0.9565	0.9871

ATTN						
CNN-BiLSTM-ATTN	0.9667	0.8610	0.8703	0.8616	0.9701	0.9865
ResNet1D	0.9736	0.8496	0.9056	0.8747	0.9744	0.9914
HMR-BiLSTM (Proposed)	0.9749	0.8562	0.9146	0.8825	0.9749	0.9916

As detailed in Table 1, classical baselines struggle with the severe class imbalance. The classical methods fall apart on the minority classes (Logistic Regression macro F1 0.4787, Decision Tree 0.6846), while every neural model holds macro F1 above 0.86 and AUC above 0.98. This supports H1: sequential and convolutional networks fit ECG classification better than traditional machine learning.

Among the neural models, HMR-BiLSTM takes the top macro recall (0.9146), macro F1 (0.8825), and AUC (0.9916). Against CNN-BiLSTM-ATTN it lifts recall by 4.43 pp and macro F1 from 0.8616 to 0.8825. ResNet1D is a strong modern CNN baseline, beating both recurrent attention models on accuracy and AUC (0.9736 accuracy, 0.9914 AUC), but HMR-BiLSTM still edges past it on the two metrics that matter for rare beats, macro F1 by 0.78 pp and macro recall by 0.90 pp. That fits what the recurrent design is for: rare classes need temporal context across the whole heartbeat. Precision slips slightly (0.8562 against CNN-BiLSTM-ATTN's 0.8610), but it comes with higher recall, the trade you want in cardiac monitoring where a missed arrhythmia costs more than a false alarm. The per-class numbers show where the gains land: V, F, and Q reach high AUC (0.9972, 0.9971, 0.9995), and F1 hits 0.9492 for V and 0.7944 for F, both above CNN-BiLSTM-ATTN. S stays the hardest case (0.9698 AUC, 0.7387 F1), since supraventricular beats look almost like normal sinus rhythm without RR-interval context.

5.2 Ablation Study

We ablate six components of HMR-BiLSTM to isolate their individual contributions. **Table 2** and **Table 3** report two groups of ablations trained under identical hyperparameters; each group is compared against its own full-model baseline due to run-to-run variation inherent in GPU non-deterministic operations.

Table 2: . Architectural ablation on the MIT-BIH test set. Baseline: full HMR-BiLSTM (F1 = 0.8921).

Variant	Params	F1-macro	F1-S	F1-V	F1-F	AUC
HMR-BiLSTM (full)	505,038	0.8921	0.7387	0.9492	0.7944	0.9916
No-RMC ($c_t = c_{lstm}$)	505,038	0.891	0.759	0.950	0.771	0.991
No-CNN (raw input)	450,062	0.847	0.684	0.932	0.666	0.988
Mean-	486,41	0.850	0.642	0.926	0.721	0.986

Pool (no attention)	3	6	7	1	6	5
No-Interaction ($m_t = 0$)	8	3	1	6	7	2

Table 3: Training objective and pathway ablation. Baseline: full HMR-BiLSTM (F1 = 0.8804, separate training run).

Variant	Params	F1-mac	F1-S	F1-V	F1-F	AUC
HMR-BiLSTM (full)	505,038	0.8804	0.7179	0.9338	0.8037	0.9885
No-Hybrid-Path ($c_t = c_{rmc}$)	505,038	0.8606	0.6307	0.9338	0.7720	0.9854
No-Smoothness ($\lambda=0$)	505,038	0.8919	0.7424	0.9540	0.7831	0.9921
No-Adv-Training	505,038	0.9023	0.7789	0.9509	0.8025	0.9933

CNN frontend and attention pooling: Removing the CNN causes the largest single drop in macro F1 (0.8921 $\rightarrow$ 0.8470, -4.5 pp) and F1-F collapses by 12.8 points. Without convolution the recurrent block receives raw input at $T = 187$ steps and cannot recover localised morphological features like QRS shape or P/T peaks. Swapping attention pooling for mean pooling costs 4.2 points on macro F1 and hits S-beats hardest (-9.6 pp). S-beats differ from Normal in a narrow time window that averaging dilutes; attention can focus on exactly that window.

RMC pathway: Turning off the whole RMC pathway (No-RMC) barely moves macro F1 (0.8918 vs 0.8921), but F1-F drops from 0.7944 to 0.7713. Fusion beats are only 0.73% of the data, so a 2.3-point fall there vanishes in the macro number. That average hides the part that matters in the clinic. In ambulatory monitoring, even a few caught Fusion events can point to a mixed-origin arrhythmia. No-RMC also strips out the model’s internal gate signal entirely. So the pathway does nothing for the headline score, yet it holds up the one class where a miss costs the most.

Interaction term: To isolate the multiplicative interaction m_t , we train a No-Interaction variant that sets $m_t = 0$ and keeps the rest of the RMC pathway intact. F1-F falls by 1.5 points (0.7944 $\rightarrow$ 0.7797) and F1-S by 3.8 points (0.7387 $\rightarrow$ 0.7011), so the interaction alone accounts for much of what RMC contributes on the Fusion class. The S-class result is the telling one. Removing the whole RMC pathway leaves S-beat detection untouched, yet removing only m_t hurts it. The element-wise product of cell state and hidden state therefore seems to encode something the model uses

specifically to separate Supraventricular beats, although this ablation does not pin down exactly what that is.

The No-Interaction variant also exhibited training instability absent from all other variants: epoch training time fluctuated from 1,076 s to 29,874 s and validation F1 oscillated between 0.66 and 0.86 before recovering. We hypothesise that this instability arises because the purely linear input $W_c \cdot c + W_h \cdot h$ to LayerNorm is more susceptible to directional collapse across the 96 hidden dimensions than the nonlinear input that includes m_t ; confirming this would require logging the LayerNorm standard deviation during training, which we leave for future work.

Hybrid Path: Removing the Hybrid Path and forcing $c_t = c_{rmc}$ drops macro F1 from 0.8804 to 0.8606, with F1-S falling from 0.7179 to 0.6307 (−8.7 pp). The LSTM backup path stabilises minority-class learning during early training when the blend gate $\beta \approx 0.007$ keeps the RMC branch nearly shut off. Without this fallback, the model must rely on the RMC pathway from the first epoch, before the keep/add decomposition has learned useful routing.

Smoothness penalty: Disabling the smoothness penalty ($\lambda = 0$) leaves aggregate metrics nearly unchanged (F1: 0.8919 vs 0.8804) but slightly reduces F1 on Fusion beats (0.8037 $\rightarrow$ 0.7831). The regulariser stabilises temporal memory behaviour rather than boosting clean-data classification directly.

Adversarial training: Removing adversarial training raises clean macro F1 to 0.9023 since the adversarial objective competes with the classification objective during optimisation. The cost appears under attack: FGSM F1 at $\varepsilon = 0.02$ drops from 0.8425 to 0.8045, the largest single-component robustness degradation in the ablation. This strongly suggests that adversarial training is a major source of the robustness margin reported in Section 5.3, though a precise decomposition would require evaluating adversarial robustness for all variants under identical training conditions. The architecture contributes minority-class recall and memory routing; adversarial training contributes robustness. Neither alone reproduces the full model's profile.

5.3 Adversarial Robustness under FGSM and PGD

Strong clean-data performance does not necessarily imply robustness under perturbation. We tested adversarial robustness with FGSM (single-step) and PGD-20 (20 iterations, step size $\alpha = 0.005$, random start, L_∞ projection). Perturbation budgets are $\varepsilon \in \{0, 0.01, 0.02, 0.05\}$ on Z-score normalised inputs. At $\varepsilon = 0.02$ the perturbed signal is visually indistinguishable from the original, making robustness evaluation relevant for ECG applications.

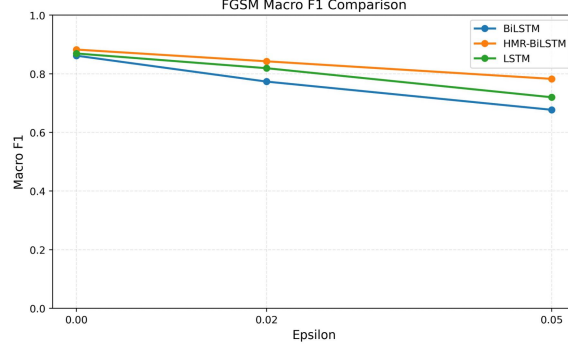


Fig. 1. Macro F1 under FGSM attack as a function of the perturbation radius ϵ . HMR-BiLSTM retains the highest F1 across all ϵ values.

Fig. 1. shows macro F1 under FGSM. At $\epsilon = 0.02$, HMR-BiLSTM stays at 0.842 versus 0.819 for CNN-LSTM-ATTN and 0.773 for CNN-BiLSTM-ATTN. At $\epsilon = 0.05$ it holds at 0.783 while CNN-BiLSTM-ATTN drops to 0.683. Under PGD-20 the ranking is the same: 0.839 at $\epsilon = 0.02$ for HMR-BiLSTM versus 0.756 for CNN-BiLSTM-ATTN, widening to 0.745 versus 0.573 at $\epsilon = 0.05$. HMR-BiLSTM retains a robustness margin under both single-step and iterative attacks. Because the proposed model is trained with adversarial examples while the baselines are not, this margin reflects the joint effect of the RMC architecture and the adversarial training procedure. The No-Adv ablation in **Table 3** isolates the training effect: without adversarial training, clean F1 rises to 0.9023 but FGSM F1 at $\epsilon=0.02$ drops to 0.8045, suggesting that the training procedure contributes more substantially to the robustness gain than the architecture alone.

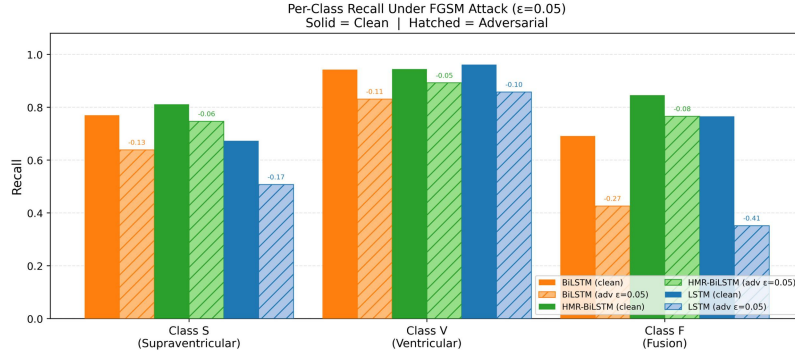


Fig. 2. Per-class recall under FGSM at $\epsilon = 0.05$. Solid = clean, hatched = adversarial. The Fusion-class recall of CNN-BiLSTM-ATTN and CNN-LSTM-ATTN (-0.27 and -0.41) while HMR-BiLSTM degrades by only -0.08 .

What matters most clinically is not aggregate F1 but per-class recall, because an aggregate number can hide the fact that the attack is selectively erasing the rare classes. **Fig. 2** shows per-class recall at $\epsilon = 0.05$ under FGSM. On the Fusion class, HMR-BiLSTM drops by only 0.08 (0.846 to 0.766), while CNN-BiLSTM-ATTN

loses 0.27 (0.691 to 0.421) and LSTM loses 0.41 (0.765 to 0.355). On S the pattern is similar: HMR-BiLSTM drops 0.06, LSTM drops 0.17. Under PGD-20 at $\varepsilon = 0.02$ the Fusion drop is 0.05 for HMR-BiLSTM versus 0.11 for both baselines. The robustness advantage concentrates on the rare, high-risk classes where HMR-BiLSTM already shows the largest clean-data gains.

5.4 Noise Robustness

For noise robustness (**Fig. 3**), all models hold steady at $\sigma \leq 0.05$. From $\sigma = 0.3$ the separation becomes clear: HMR-BiLSTM keeps F1 at 0.4798 while CNN-BiLSTM-ATTN and CNN-LSTM-ATTN collapse to 0.3451 and 0.3507. At $\sigma = 0.4$ the margin grows to 0.3620 versus 0.2386 for CNN-BiLSTM-ATTN. Normally a more complex architecture degrades faster under heavy noise, not slower. HMR-BiLSTM inverts this pattern, which may reflect implicit input smoothing by the residual gate, though we do not test this directly. From a security standpoint, gradual degradation is harder to silently exploit than a sudden collapse at some noise threshold.

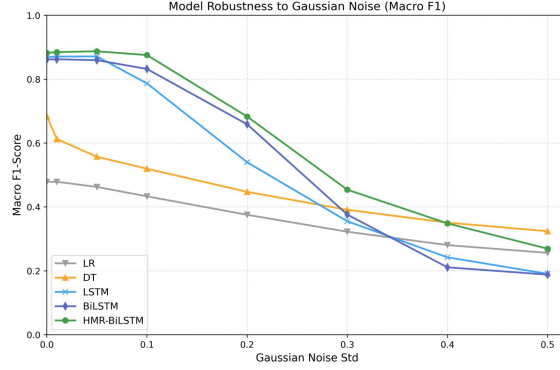


Fig. 3. Macro F1 of all five methods as a function of the additive Gaussian noise standard deviation σ . HMR-BiLSTM is the most noise-tolerant model from $\sigma \geq 0.3$.

5.5 Calibration

HMR-BiLSTM has a higher ECE than the baselines (0.0397 vs 0.0056 for CNN-LSTM-ATTN and 0.0080 for CNN-BiLSTM-ATTN). The low ECE of CNN-LSTM-ATTN should be read with caution: top-label ECE on an imbalanced dataset is dominated by the majority class, and a model that predicts N with high confidence on 82% of samples will appear well-calibrated even if its minority-class probabilities are unreliable. Brier score, which penalises the full probability vector rather than only the top label, reverses the ranking: 0.0444 for HMR-BiLSTM versus 0.0461 and 0.0542. The higher ECE of HMR-BiLSTM likely reflects the combined effect of Focal Loss (which reshapes the probability landscape away from standard cross-entropy calibration) and adversarial training (which broadens the predictive distribution for

robustness). Neither metric alone captures calibration quality on a 5-class imbalanced task; classwise ECE would be more informative but is not reported here. We do not claim a calibration advantage for HMR-BiLSTM — the evidence is mixed.

5.6 Interpretability Analysis

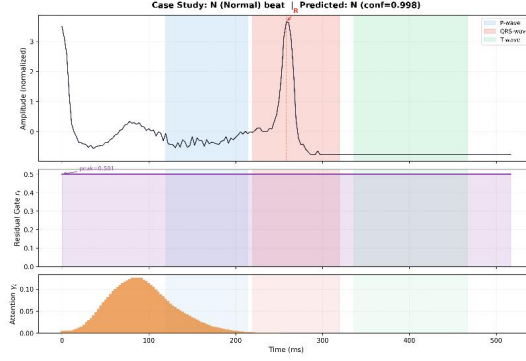


Figure 4: Case Study Normal with the raw ECG waveform, estimated P/QRS/T regions and attention pooling weights γ_t on a shared time axis

Figure 4 shows that the residual gate r_t averages to approximately 0.500 across time steps due to the LayerNorm–sigmoid architecture and does not vary at the individual-beat level; its contribution is a memory retention bias analysed in Section 5.2. The proposed clinician-facing interpretability is primarily derived from these attention weights.

Qualitative observation suggests that the attention distribution shifts in a manner consistent with clinical expectations across beat types. For Normal beats, attention peaks before the QRS complex — the P-wave and early QRS region that defines normal sinus rhythm. For Ventricular beats, attention shifts to the post-QRS T-wave region, where the wide QRS and ST-deviation that characterise ventricular ectopy are most visible. For Fusion beats, attention concentrates early (~ 60 ms) at the pacemaker spike, where intrinsic and paced QRS overlap. For Supraventricular beats, attention splits into two peaks, one near the P-wave and one near the QRS, which tracks the subtle PR-interval differences that set this class apart from Normal.

In practice, a cardiologist checking a flagged beat could read the attention distribution directly off the ECG trace, seeing which part of the waveform drove the model's decision. If the model flags a beat as Ventricular and the attention peak sits on a widened QRS region, the flag appears consistent with known pathology, potentially streamlining the clinician's review process. Conversely, if the attention peak sits on a baseline segment with no morphological abnormality, it serves as a visual cue for the clinician to question the flag. The attention overlay does not replace clinical judgement but is designed to potentially reduce the inspection effort by guiding the clinician to check whether the highlighted region matches the expected abnormality rather than exhaustively reading the entire waveform. However, formal

validation by clinical experts has not yet been conducted and is reserved for future work.

6 Discussion and Conclusion

On MIT-BIH, HMR-BiLSTM reaches macro F1 0.8825, recall 0.9146, and AUC 0.9916, ahead of CNN-BiLSTM-ATTN by 2.09 pp on F1 and 4.43 pp on recall. Most of this gain comes from the Supraventricular and Fusion classes, which are clinically significant and difficult for standard LSTM. Under adversarial perturbation at $\epsilon=0.02$, macro F1 stays at 0.842 (FGSM) and 0.839 (PGD-20) versus 0.773 and 0.756 for CNN-BiLSTM-ATTN, with the gap widening at $\epsilon=0.05$. The same ordering holds for Gaussian noise beyond $\sigma=0.3$, and Brier score also favours HMR-BiLSTM (0.0444 vs 0.0461 and 0.0542). Overall, the proposed framework demonstrates improved robustness compared with the evaluated recurrent baselines — but we want to be explicit that none of this amounts to a deployment-ready claim.

What this paper establishes is narrow but concrete. The residual gate mechanism lifts recall on the minority classes and holds up better than the baselines we tested when the input is perturbed, both by adversarial attacks and by ordinary noise. Alongside it, the attention weights settle on the parts of the waveform that carry diagnostic meaning, so a clinician can lay them over the raw ECG and see what the model attended to. The gate itself plays a quieter role, biasing how memory is routed inside the cell rather than producing a trajectory anyone can read off directly.

Several limitations should be acknowledged. First, the experiments were conducted on the intra-patient Kaggle version of MIT-BIH rather than the inter-patient AAMI EC57 protocol [21], which may lead to optimistic performance estimates. Second, classification performance on the S and F categories remains below that of the dominant Normal class despite the use of class-balancing strategies. Third, privacy-preserving techniques such as federated learning, DP-SGD, and secure aggregation were not considered in the current study. Stronger adaptive attacks such as Carlini–Wagner and AutoAttack, and cross-dataset validation on Vietnamese clinical ECG, remain for future work.

Acknowledgement

This work was supported by the University of Economics Ho Chi Minh City (UEH), Vietnam.

Data Availability

The ECG datasets used in this study are publicly available from the MIT-BIH Arrhythmia Database. The implementation code, experimental configurations, and evaluation results are available at: <https://github.com/tn1-ai/HMR-BiLSTM.git>

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